

Elio Govea

Senior Software Engineer · [Vaisala](#)

Systems engineer in networking and embedded Linux, now building cross-platform systems on Linux and Windows — from home routers to aviation weather platforms. ICPC World Finalist.

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Summary

Eight years of production C++ across networking, embedded Linux and cross-platform delivery. At [Vaisala](#), extending [AviMet](#) — an aviation weather platform running in 170+ countries — onto new Linux targets, and building the simulation tooling its modernization is verified against. Before that, four and a half years at [F-Secure](#) on the Sense SDK: home-router security embedded by 200+ service-provider partners and shipped across x86, ARM and MIPS, where I traced a production cost problem to its root and cut **monthly CDN spend by 80%**.

Skills

LANGUAGES	C++ (11–20), C, Python, Bash, PowerShell, TypeScript / JavaScript
NETWORKING	netlink, netfilter, iptables, nfqueue, nflog, conntrack, BPF, Unix domain sockets
PROTOCOLS	IPv4 / IPv6, TCP, UDP, DHCP, DNS, HTTP / HTTPS, TLS, QUIC
PLATFORMS	Embedded Linux, Buildroot, OpenWrt, Alma Linux, x86 / ARM / MIPS, Windows
BUILD & CI	CMake, GNU Make, Cross-compilation, Git, Docker, LXC, Jenkins, AWS CI
FRAMEWORKS	Qt, Robot Framework, OpenGL
CONCEPTS	Algorithms & data structures, Performance optimization, Static & dynamic analysis, Debugging & profiling

Experience

[Vaisala](#) · Senior Software Engineer Feb 2025 – Present
Vantaa, Finland · [AviMet](#) aviation weather platform — deployed across 170+ countries, 2,500+ systems delivered

- Made AviMet's behavior provable to the standard a regulated aviation domain demands — one programmable reference driving both the legacy and modernized systems into arbitrary states from simulated sensor data, with time-varying weather scenarios and deterministic setup, so correctness can be verified repeatably and equivalence testing becomes possible.
- Replaced GUI-bound tooling with an interface developers, testers and AI-assisted workflows can script — what previously meant working through an extensive configuration checklist by hand now runs from defaults almost immediately, turning scattered team knowledge into executable documentation.
- Led AviMet's expansion onto **Alma Linux** — extending the C++/Qt/CMake build matrix and toolchain to a second Linux distribution, and implementing Linux-side license enforcement against Vaisala's existing licensing system.
- Built tooling for a new cloud entitlements system, designed for reuse across Vaisala products beyond AviMet.
- Hardened a long-lived codebase as it moved onto new platforms — root-causing memory leaks and corner-case crashes latent for years.

[F-Secure](#) Sep 2020 – Feb 2025
Helsinki, Finland · Poznań, Poland · Sense SDK — C++ home-router security embedded by 200+ service-provider partners

Senior Software Engineer May 2023 – Feb 2025

- Cut **monthly CDN spend by 80%** with no user-visible impact. Root-caused it to router-side behavior driving excessive traffic; a router update would have taken months to reach deployed devices, so I shipped a backend-side mitigation in days instead, then landed the router-side fix for future releases to close the cause permanently.

- Co-designed and implemented cloud-driven dynamic traffic-rule enforcement, letting F-Secure push new security policy to deployed routers without firmware updates.

Software Engineer

Sep 2020 – Apr 2023

- Built core Sense SDK functionality — URL filtering and parental controls (forcing safe-mode browser engines), traffic-statistics collection, device recognition and fingerprinting — across IPv4/IPv6, DHCP, TCP/UDP, DNS and HTTP/HTTPS, using Linux networking interfaces (netlink, netfilter, iptables, nfqueue, nflog, conntrack, BPF) against cloud back-end services.
- Built the embedded Linux clients consuming those cloud services, the Unix-domain-socket IPC between clients and services, and reworked the security-event notification path.
- Owned the cross-platform build system (CMake + GNU Make) for fast, reproducible builds across x86, ARM and MIPS toolchains, plus SDK integrations for reference platforms that shortened partner demos and onboarding.
- Designed the Jenkins/AWS CI pipelines and the quality gates behind them — coverage, static and dynamic analysis — producing builds for every supported architecture and reference platform.
- Built the test infrastructure in Python and Robot Framework, including recovery paths for OS- and cloud-service-level failures.

Universidad de Pinar del Río · Software Engineer

Jan 2018 – Aug 2020

Pinar del Río, Cuba

- Built real-time C++/Qt applications for vibration-signal processing and IMU-based object position tracking — embedded sensor input over serial (Arduino), Linux, OpenGL visualization.
- Served as ICPC judge for the Caribbean region and coached a team to the ICPC Latin American Finals, alongside teaching C++, Python, Arduino, Linux and Raspberry Pi.

Education

Universidad de Pinar del Río · B.Sc. Telecommunications and Electronics Engineering

Sep 2012 – Jul 2017

Pinar del Río, Cuba · GPA 4.95 / 5.00 · ICPC World Finalist

Achievements

- **Honorable Mention** — 2018 [ICPC World Finals](#), Beijing (*140-team final, from ~50,000 regional competitors*)
- **2nd Place, Silver Medal** — 2016 ICPC Latin American Finals
- **4th Place** — 2015 ICPC Latin American Finals
- **Round 3, Top 1000** — 2019 Google Code Jam (*of ~50,000 entrants — top ~2%*)

Selected projects

DNS library — Wire format parsed and generated from scratch.

C++ project template — Identical builds on any machine.

Baseball Limits 2D — Pareto frontiers over 150 years of MLB data, shipped client-side. [Live demo ↗](#)

Router-network Docker testbed — Client → router → server, in containers, no hardware.

Automatic layout for a growing DAG — Construction order fixes one axis; forces settle the rest. [Live demo ↗](#)

[More projects →](#)